

# Trustworthy Cardiac Digital Twins: An Intended-Use Assurance Framework for Multimodal Modeling and Clinical Translation

Weihaio Cheng, Huihui Ye, and Huafeng Liu

**Abstract**—Cardiac digital twins integrate patient-specific measurements with mechanistic, data-driven, or hybrid models to estimate cardiovascular state, test interventions, and update predictions as new data arrive. Their clinical promise is clearest when they answer a bounded decision that static risk scores or isolated artificial-intelligence models cannot address. Evidence of readiness, however, varies sharply across intended uses. This state-of-the-art review maps literature through July 2026 across multimodal data, model personalization, clinical applications, and translational assurance. Recent feasibility evidence is strongest in ventricular tachycardia ablation, whereas atrial fibrillation, heart failure, coronary and structural intervention, congenital disease, remote monitoring, and in silico trials remain at heterogeneous stages of technical and clinical validation. To separate model sophistication from clinical readiness, we introduce the Cardiac Digital Twin Assurance Matrix (CDT-AM). The matrix links intended use and failure consequence to six dimensions: data provenance and quality; model verification and parameter identifiability; uncertainty and calibration; clinical validity and utility; human factors and equity; and lifecycle governance. A complementary evidence ladder distinguishes conceptual and in silico work from retrospective validation, prospective workflow feasibility, comparative clinical impact, and monitored deployment. We conclude that trustworthy translation requires evidence for the complete data-to-decision pathway, including uncertainty-aware outputs, workflow evaluation, human oversight, and controlled updating. This intended-use framework provides biomedical engineers and clinicians with a practical basis for evaluating what a cardiac digital twin has demonstrated and what evidence remains necessary before it can influence care.

**Index Terms**—Cardiac digital twin, precision cardiology, mechanistic modeling, multimodal data, ventricular tachycardia, uncertainty quantification, clinical translation.

## I. INTRODUCTION

CARDIOVASCULAR decisions are made under pronounced patient heterogeneity. Scar architecture, conduction, ventricular loading, comorbidity, disease trajectory, and treatment response can differ substantially among patients who share a diagnosis [1]–[3]. Population risk models remain indispensable, but they do not directly test how an individ-

ual heart may respond to an ablation lesion, device setting, revascularization strategy, or change in loading.

A cardiac digital twin (CDT) is a patient-specific computational representation of the heart or cardiovascular system that combines individual measurements with mechanistic, data-driven, or hybrid models to estimate state, simulate mechanisms, predict trajectories, or compare interventions [4]–[6]. Patient specificity alone is insufficient. A clinically meaningful twin also requires a defined physical counterpart, an intended use, explicit data-to-model coupling, uncertainty-aware inference, and a pathway for updating or feedback. These properties distinguish a CDT from a static risk score, a single-task artificial intelligence (AI) predictor, an offline virtual-heart model, or a population-level in silico trial.

Translation has accelerated. Personalized electrophysiology models have prospectively identified VT substrate and informed virtual ablation [7], [8]. A 10-patient feasibility study subsequently reported digital-twin-guided VT ablation, with noninducibility in all participants and no sustained recurrence in eight during a mean 13-month follow-up [9]. These uncontrolled findings are encouraging but do not establish comparative effectiveness. The 60-participant APOLO-VT investigation is designed to provide a larger prospective assessment of a multimodal noninvasive-to-invasive workflow [10].

Two recent evidence reviews show why a further synthesis is needed. A systematic review of 42 cardiovascular digital-twin studies and a cardiovascular scoping review found heterogeneous definitions, limited external or prospective validation, and sparse implementation reporting [11], [12]. Broader reviews have catalogued circulatory models and digital-twin components, while a recent perspective centers on verification, validation, and uncertainty quantification [13], [14]. These contributions describe modalities, models, applications, and credibility methods well, but they do not consistently connect a proposed clinical decision and its failure consequences to the complete evidence required for use. This review therefore asks three questions: (1) which data and models are necessary for different CDT functions; (2) what level of evidence exists for each clinical intended use; and (3) which assurance claims must be satisfied before a CDT can influence care? Fig. 1 summarizes the data-to-decision pathway. Our principal contribution is an intended-use assurance framework that separates technical performance from clinical readiness and makes the missing evidence for a bounded claim explicit.

Weihaio Cheng is with the School of Communication Engineering, Hangzhou Dianzi University, Hangzhou 310018, China (e-mail: weihaio\_cheng@hdu.edu.cn).

Huihui Ye is with the School of Communication Engineering, Hangzhou Dianzi University, Hangzhou 310018, China (e-mail: yehuihui@hdu.edu.cn). Corresponding author: Huihui Ye.

Huafeng Liu is with the Key Laboratory for Biomedical Engineering of Ministry of Education, College of Biomedical Engineering and Instrument Science, Zhejiang University, Hangzhou 310027, China (e-mail: liuhf@zju.edu.cn).

## II. REVIEW METHOD AND ANALYTICAL FRAMEWORK

### A. Search and Evidence Mapping

This article is a structured state-of-the-art review, not a formal systematic review or meta-analysis. Searches were updated through July 19, 2026 using PubMed, OpenAlex, Crossref, and publisher databases, with targeted checks in IEEE Xplore and citation tracing. Search concepts combined “cardiac” or “cardiovascular” with “digital twin”, “virtual heart”, “patient-specific model”, “computational electrophysiology”, “ventricular tachycardia”, “atrial fibrillation”, “heart failure”, “coronary”, “structural heart”, “in silico trial”, “verification”, “validation”, and “uncertainty quantification”. We prioritized peer-reviewed primary studies, recent evidence reviews, and authoritative standards or regulatory documents. Seminal pre-2010 modeling studies were retained when needed for technical foundations.

Studies were included when they contributed to patient-specific cardiovascular representation, multimodal personalization, simulation, counterfactual therapy testing, longitudinal updating, validation, or deployment. Generic cardiovascular predictors were considered enabling modules rather than twins unless they were embedded in a patient-specific simulation or updating loop. For clinical claims, primary studies were preferred over reviews. Each retained reference was checked against DOI, PubMed, publisher, Crossref, OpenAlex, or official-agency metadata; title–DOI mismatches were corrected during revision. Because the literature is heterogeneous and no pooled effect was estimated, maturity labels in this review are interpretive evidence categories rather than quantitative grades.

### B. Intended Use and Evidence Ladder

Assessment begins with the intended user, target population, decision, time horizon, comparator, and plausible harm from error. We use a six-level evidence ladder: E0, concept or architecture; E1, in silico verification; E2, retrospective or external technical validation; E3, prospective workflow feasibility; E4, comparative clinical-impact evaluation; and E5, monitored post-deployment performance. An application can occupy different levels for different claims. For example, a VT twin may reach E3 for workflow feasibility while remaining below E4 for recurrence reduction. This distinction prevents a technically validated component from being treated as a clinically validated system.

Throughout this review, verification, validation, and uncertainty quantification (VVUQ) refer to the linked processes used to establish that software solves the intended model, model outputs are credible for a specified context of use, and relevant uncertainty is characterized and propagated.

## III. CONCEPTUAL FOUNDATIONS

Digital twins inherit engineering concepts of a virtual representation coupled to a physical counterpart, but cardiovascular usage spans a continuum from offline models to continuously updated decision systems [6], [13], [15]. The field also builds on the Virtual Physiological Human, whole-heart simulation,

and patient-specific cardiovascular modeling [1], [3], [16]. Table I makes the coupling level explicit.

A patient-specific model may be clinically useful without being a full digital twin. Conversely, adding streaming data does not make a system clinically credible. The label should be reserved for systems whose patient linkage and updating are material to the intended function. Static risk scores and single-task AI modules can support a twin, for example by segmenting images or estimating parameters, but they do not independently provide counterfactual simulation or closed-loop updating. Virtual cohorts and in silico trials address population-level questions and should not be conflated with individual twins [17], [18].

The operational distinction determines what must be validated. A research model that explains reentry requires credible equations and sensitivity analysis. An ablation-planning twin additionally requires registration accuracy, target reproducibility, integration with mapping systems, and prospective workflow evaluation. A longitudinal heart-failure twin must also manage missingness, device changes, drift, alert thresholds, and recalibration. Authors should therefore report the user, population, decision, update frequency, comparator, failure modes, and human override before reporting aggregate performance.

## IV. MULTIMODAL DATA FOUNDATIONS

Data quality sets the practical ceiling for digital-twin fidelity. Fig. 2 organizes the major data streams, and Table II summarizes their distinct roles. Cardiovascular magnetic resonance (CMR), computed tomography (CT), echocardiography, nuclear imaging, and 4D-flow CMR provide anatomy, motion, perfusion, tissue characterization, and flow, including phenotyping in coronary artery disease (CAD) [19]–[23]. Reconstruction and segmentation are therefore part of the measurement chain, not neutral preprocessing steps: acceleration, resolution, contour uncertainty, and registration can alter personalized parameters [24], [25]. Late-gadolinium-enhancement CMR is particularly important for scar-related VT modeling, but scar thresholds and image-to-mesh transfer remain sources of target uncertainty [26]–[28].

Electrophysiological data anchor twins designed for arrhythmia analysis, including atrial fibrillation (AF), sudden cardiac death (SCD) risk, and ablation planning. The 12-lead ECG, Holter monitoring, wearable ECG, intracardiac electrograms (EGMs), and electroanatomical mapping characterize activation, repolarization, conduction delay, rhythm burden, and arrhythmogenic substrate [29], [30]. Surface signals are scalable and longitudinal, whereas invasive mapping is spatially detailed but procedure-limited. Both require careful synchronization, signal quality assessment, and interpretation under uncertainty.

Hemodynamic and biomechanical data are central for heart failure (HF), valvular disease, congenital heart disease (CHD), pulmonary hypertension, and device optimization. Blood pressure (BP), Doppler echocardiography, catheter pressures, pressure–volume loops, 4D-flow CMR, ventricular strain, and biomarkers constrain loading, contractility, vascular resistance,

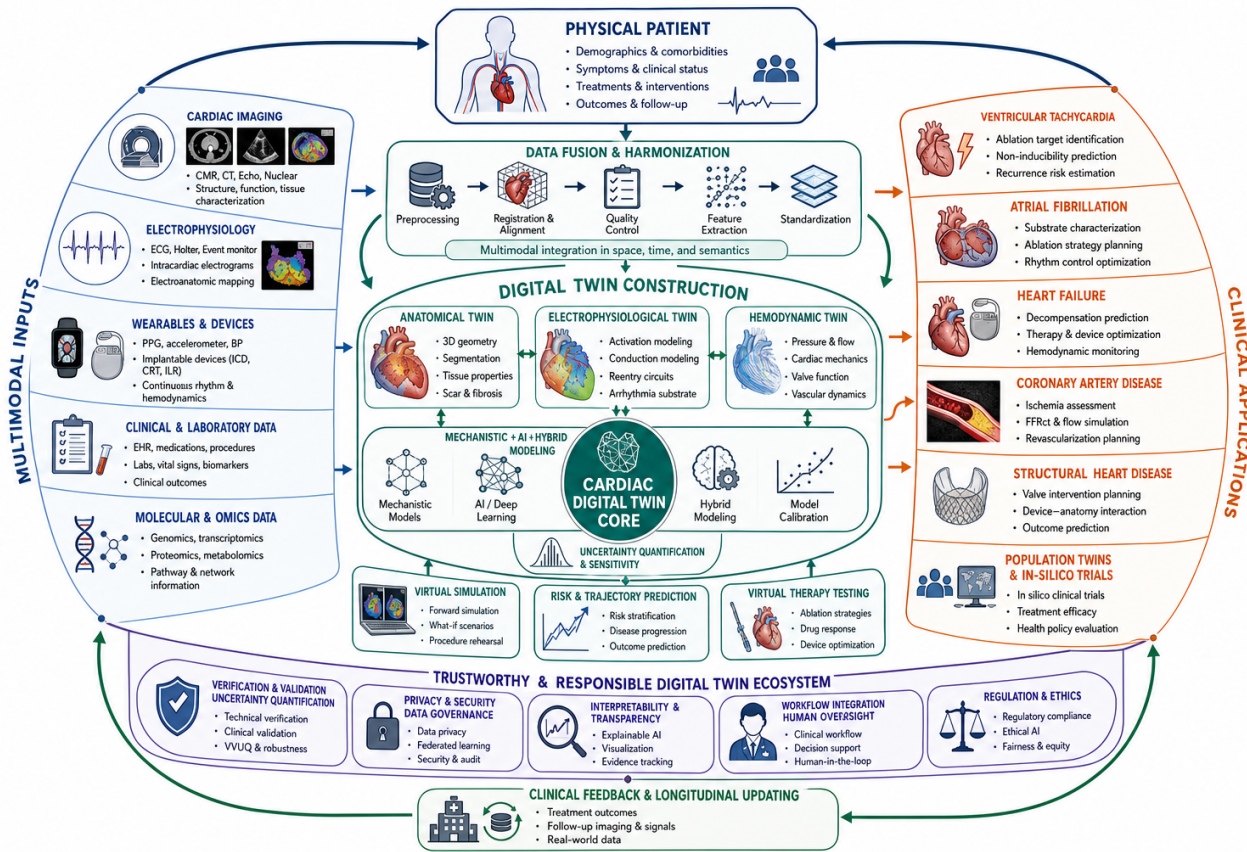


Fig. 1. Conceptual framework of cardiac digital twins for precision cardiovascular medicine. Multimodal patient data are harmonized to construct patient-specific anatomical, electrophysiological, hemodynamic, and clinical representations. Mechanistic, AI, and hybrid models support simulation, prediction, virtual therapy testing, uncertainty quantification, and clinical decision support. Translation requires verification, validation, privacy governance, interpretability, workflow integration, and longitudinal feedback.

TABLE I  
OPERATIONAL CONTINUUM FROM CARDIOVASCULAR DIGITAL MODELS TO CLINICALLY ACTIONABLE TWINS.

| Representation                    | Patient coupling                                                        | Typical function                                                       | Evidence boundary                                                                         |
|-----------------------------------|-------------------------------------------------------------------------|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|
| <b>Digital model</b>              | No live patient linkage; population or exemplar parameters.             | Mechanistic study, virtual experiment, method development.             | Validity is limited to stated assumptions and simulated conditions.                       |
| <b>Patient-specific model</b>     | Calibrated once from individual measurements.                           | Anatomy reconstruction, state estimation, procedural simulation.       | Personalization does not establish longitudinal validity or clinical utility.             |
| <b>Digital shadow</b>             | One-way physical-to-virtual updates.                                    | Monitoring, trajectory estimation, recalibration.                      | Updates may improve state tracking but do not prove safe decision support.                |
| <b>Cardiac digital twin</b>       | Repeated patient updates plus simulation of candidate actions.          | Counterfactual testing, planning, prediction, longitudinal management. | Requires end-to-end VVUQ, workflow evaluation, and bounded intended use.                  |
| <b>Clinically actionable twin</b> | Twin output is integrated into a supervised care pathway with feedback. | Decision support that can change management.                           | Requires prospective utility, human-factors safety, governance, and lifecycle monitoring. |

wall stress, and circulatory reserve [31]–[33]. Because several clinically important parameters are latent or only weakly identifiable, personalization should report priors, calibration targets, and parameter correlations rather than a single best-fit state [34]–[36].

Wearables, home monitoring, and implantable devices enable longitudinal updating. Smartwatch photoplethysmography, wearable ECG, activity, sleep, oxygen saturation, home BP, pacemaker diagnostics, implantable cardioverter-defibrillator (ICD) events, and cardiac resynchronization ther-

apy (CRT) data can detect deviations from baseline, quantify rhythm burden, and support early warning [37], [38], [39]. Yet these streams also introduce artifacts, missingness, device heterogeneity, user adherence problems, and alert fatigue. For clinical deployment, digital twins must distinguish clinically meaningful change from noise.

Electronic health records (EHRs), laboratory tests, and omics provide clinical context: diagnoses, medications, procedures, reports, hospitalizations, outcomes, renal function, lipids, troponin, natriuretic peptides, inflammatory markers,

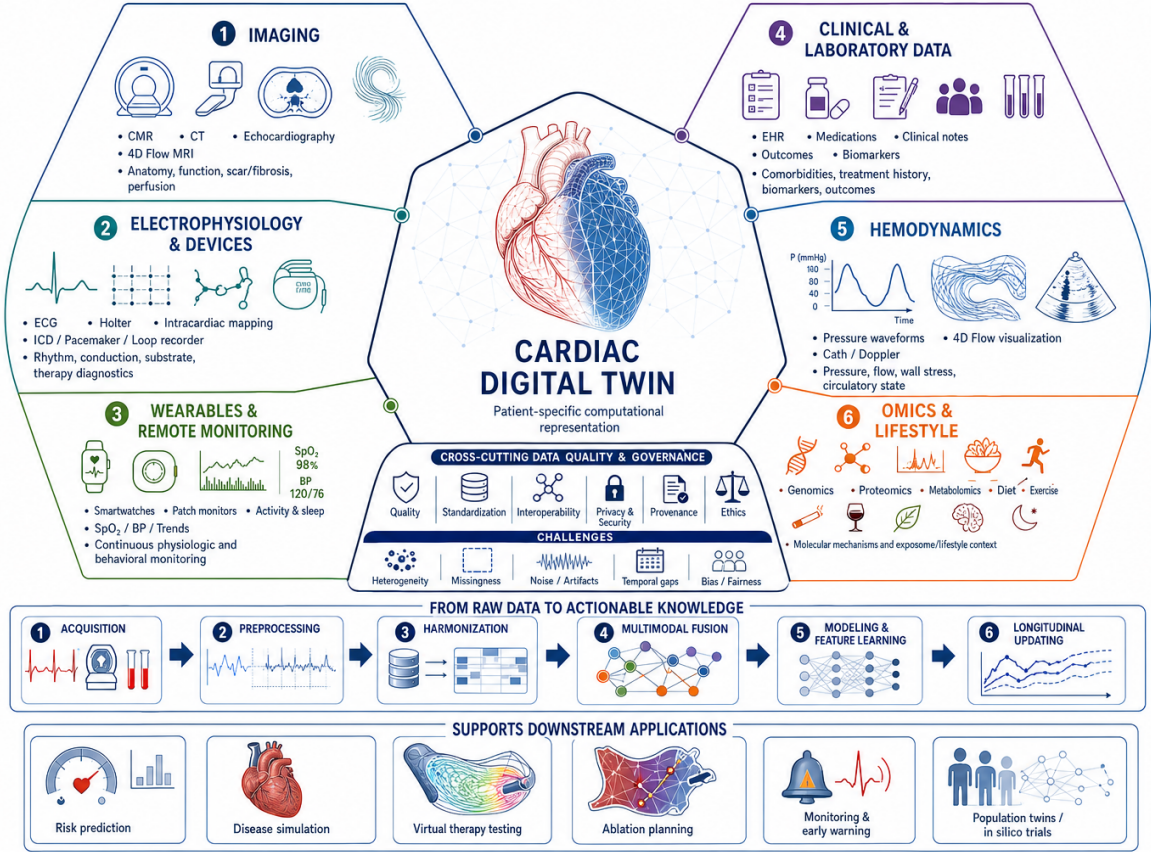


Fig. 2. Multimodal data ecosystem for cardiac digital twins. Imaging, electrocardiography (ECG) and intracardiac signals, hemodynamics, wearables, implantable devices, laboratory tests, electronic health records, omics, and outcomes provide complementary views of patient state. Their clinical value depends on harmonization, quality control, interoperability, privacy protection, and longitudinal updating.

and inherited risk [40], [41]. Natural-language processing may structure notes and reports [42], [43], but every extracted datum used for personalization should remain traceable to its source because an extraction error can propagate through simulation and decision support.

Across all modalities, harmonization is a major bottleneck. Imaging may reside in DICOM archives, ECGs in device-specific formats, EHRs in structured and unstructured fields, and wearables in proprietary platforms. Interoperability frameworks such as HL7 FHIR and SMART-on-FHIR, common data models, standardized annotations, and FAIR data principles are therefore essential [44], [45], [46]. Privacy-preserving architectures, including federated learning, secure computation, de-identification, edge processing, access control, and audit trails, will be critical for multicenter and longitudinal digital twins [47], [48], [49], [50], [51].

Data representativeness is equally important. Digital twins developed using single-center imaging protocols, selected device populations, or narrow demographic cohorts may not generalize to community hospitals, pediatric patients, under-represented populations, or different vendors. Data reporting should include acquisition protocols, missingness, preprocessing, segmentation uncertainty, device provenance, outcome definitions, and subgroup composition. Multimodal volume alone is not equivalent to clinical readiness unless the data are

traceable, representative, and aligned with the intended use.

## V. MODELING METHODOLOGIES

No single model class is sufficient for all cardiac digital-twin tasks. Fig. 3 shows the pipeline from measurements to simulation and feedback, while Table III compares major strategies. Mechanistic models encode ionic currents, propagation, electromechanical coupling, ventricular mechanics, vascular resistance, and circulation [2], [52]–[56]. They enable counterfactual testing, but their credibility depends on identifiability, boundary conditions, numerical verification, and sensitivity to assumptions [57].

Electrophysiological simulation is one of the most mature domains. Reaction–diffusion, monodomain, bidomain, eikonal, and reduced-order models can simulate activation, repolarization, conduction slowing, scar-mediated reentry, and ablation effects [58], [59], [60]. VT models integrate scar imaging, anatomy, tissue heterogeneity, and stimulation protocols to identify plausible circuits and targets [61], [7], [8]. Atrial fibrillation (AF) models incorporate atrial geometry, fibrosis, conduction properties, triggers, and virtual lesion sets [62], [63], [64], [65]. In both settings, clinical credibility depends on calibration against patient measurements, sensitivity analysis, and communication of alternative plausible mechanisms.

Hemodynamic and circulatory models range from lumped-parameter systems to computational fluid dynamics and fluid–

TABLE II  
CORE DATA SOURCES AND THEIR ROLES IN CARDIAC DIGITAL TWINS.

| Source                   | Examples                                     | Twin role                                                 | Applications                          | Key limitation                             |
|--------------------------|----------------------------------------------|-----------------------------------------------------------|---------------------------------------|--------------------------------------------|
| <b>Imaging</b>           | CMR, CT, echo, nuclear, 4D flow.             | Anatomy, function, scar, fibrosis, perfusion, motion.     | VT, HF, valves, CHD, CAD.             | Protocol and segmentation variability.     |
| <b>Electrophysiology</b> | ECG, Holter, wearable ECG, EGMs, mapping.    | Activation, conduction, repolarization, rhythm burden.    | VT, AF, SCD risk, monitoring.         | Noise, sparsity, temporal variability.     |
| <b>Hemodynamics</b>      | BP, Doppler, catheter pressures, PV loops.   | Loading, flow, elastance, resistance, reserve.            | HF, valves, CHD, devices.             | Latent parameters and boundary conditions. |
| <b>Longitudinal data</b> | Wearables, home BP, ICD/CRT, loop recorders. | Updating, baseline estimation, early warning.             | AF, HF, rehabilitation, chronic care. | Artifacts, adherence, vendor silos.        |
| <b>Clinical context</b>  | EHR, labs, reports, notes, outcomes, omics.  | Comorbidities, treatments, biomarkers, validation labels. | Risk, phenotyping, therapy selection. | Missingness, coding bias, privacy.         |

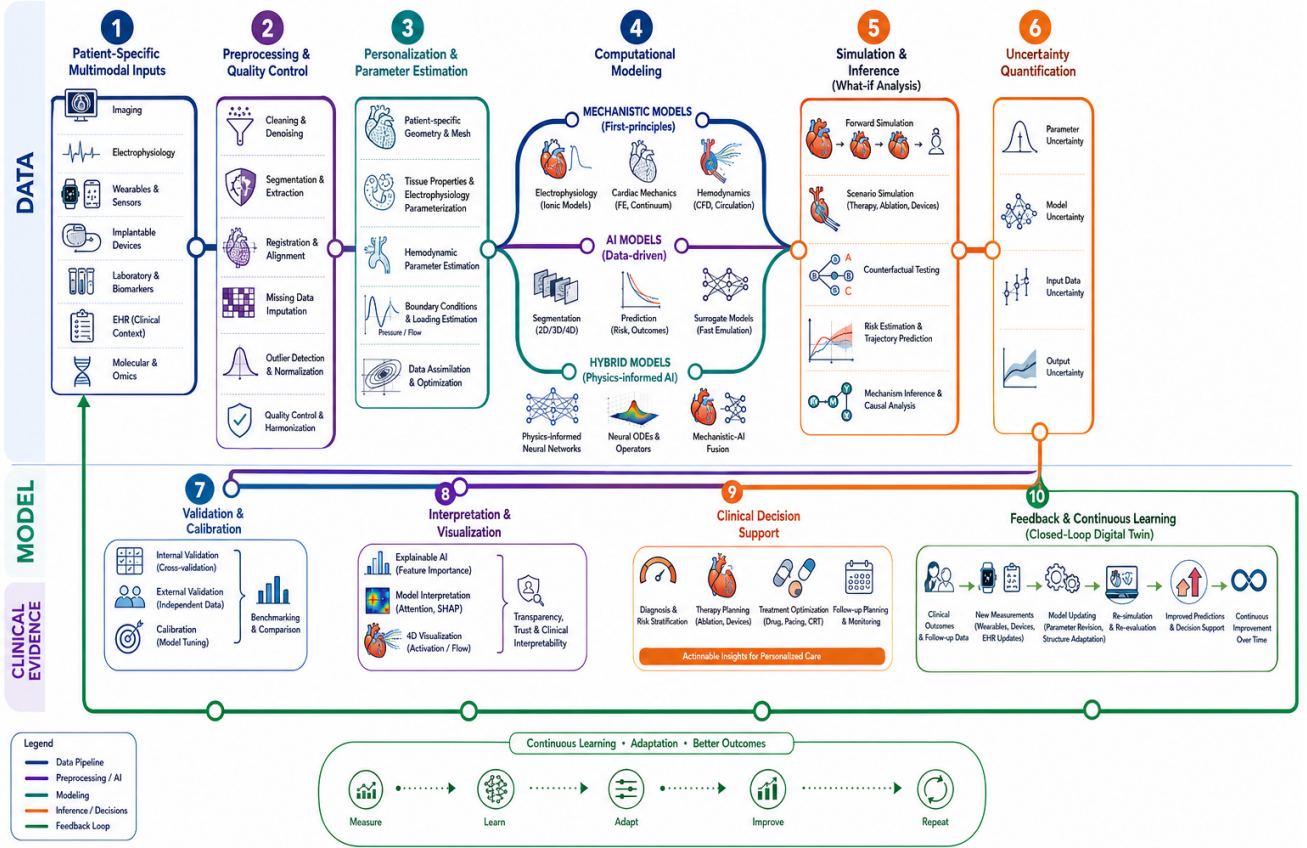


Fig. 3. Modeling pipeline for cardiac digital twins. Patient-specific data are transformed through preprocessing, personalization, mechanistic or AI modeling, simulation, uncertainty quantification, validation, interpretation, and clinical decision support. Feedback from outcomes and longitudinal monitoring enables recalibration.

structure interaction [31], [32], [33]. They are well suited to heart failure, valvular disease, congenital heart disease, pulmonary hypertension, and device optimization, where pressure, flow, stiffness, resistance, and ventricular–arterial coupling matter. Recent heart-failure work illustrates how mechanistic models can estimate latent physiological states and support interpretable patient-specific analysis [36].

Machine learning and deep learning are AI enabling tools rather than replacements for digital twins. They automate segmentation, scar or fibrosis quantification, ECG interpretation, phenotype discovery, risk prediction, parameter estima-

tion, surrogate simulation, and multimodal fusion [66]–[69]. Physics-informed learning, neural operators, sparse equation discovery, Bayesian updating, and Kalman-filter-like data assimilation can connect statistical learning with physiological constraints [70]–[73]. Hybrid mechanistic–AI models are especially promising because they combine physiological plausibility with scalable automation and rapid inference.

Personalization is an inverse problem. For measurements  $\mathbf{y}$ , simulator output  $\mathbf{g}(\boldsymbol{\theta})$ , covariance  $\boldsymbol{\Sigma}$ , and physiological regularizer  $R$ , a common estimator has the form

$$\hat{\boldsymbol{\theta}} = \arg \min_{\boldsymbol{\theta}} \|\mathbf{y} - \mathbf{g}(\boldsymbol{\theta})\|_{\boldsymbol{\Sigma}}^2 + R(\boldsymbol{\theta}). \quad (1)$$

A good fit does not guarantee that  $\theta$  is identifiable: distinct activation maps or material parameters can reproduce similar measured signals [30], [35]. Credible pipelines should therefore examine local or global sensitivity, posterior dependence, profile likelihoods, and out-of-sample quantities of interest. Sequential Bayesian or Kalman-type data assimilation can update state and parameter distributions as measurements arrive; Gaussian-process emulators, neural operators, and reduced-order surrogates can make repeated calibration and uncertainty propagation tractable, but require independent error bounds against the high-fidelity solver [72], [74]. This distinction between predictive fit, parameter recovery, and decision-relevant uncertainty is central to engineering assurance.

Foundation models and large language models (LLMs) may become bounded workflow components rather than physiological twins. Plausible roles include source-linked extraction, data-completeness checks, simulation orchestration, and clinician-facing summaries [75]–[77]. Because fluent output can conceal unsupported inference, any such component requires structured inputs and outputs, provenance, audit logs, uncertainty warnings, and human confirmation before information enters a patient-specific model.

Model choice should follow intended use rather than technological ambition. VT ablation may justify electrophysiological simulation and scar-informed anatomy; heart failure may require hemodynamic state estimation; screening may rely on AI-assisted ECG or imaging; structural interventions may need finite-element or flow simulation; and remote monitoring may require lightweight longitudinal models. A simpler, calibrated, interpretable model can be more clinically useful than a complex model that cannot be validated or deployed.

A practical cardiac digital twin is likely to be modular. One module may perform segmentation, another may estimate tissue properties, a third may simulate electrophysiology or hemodynamics, and a fourth may generate a clinician-facing report. Modularity improves engineering maintainability and allows each component to be verified separately, but it also creates interface risk: errors in segmentation, coordinate registration, parameter mapping, or unit conversion can propagate into the final recommendation. This motivates research on end-to-end uncertainty propagation, reproducible software pipelines, and clinically meaningful acceptance tests, not only better numerical solvers or larger neural networks.

## VI. CLINICAL APPLICATIONS

The most mature applications are in electrophysiology, and Table IV maps each domain to the evidence ladder. For scar-related VT, twins can reconstruct substrate, simulate reentry, localize vulnerable channels, and compare virtual lesions [7], [8], [61]. Prospective workflow evidence has now reached E3: the first digital-twin-guided ablation series enrolled 10 patients, and the 60-participant APOLO-VT study is assessing a standardized multimodal workflow [9], [10]. Neither study establishes E4 comparative benefit; recurrence, lesion burden, safety, procedure time, and performance across centers remain decisive endpoints.

AF digital twins seek to individualize rhythm control beyond uniform lesion sets. By combining atrial anatomy, fibro-

sis, conduction, rhythm data, and simulated ablation, they may help identify patients likely to benefit from pulmonary vein isolation alone versus additional substrate modification [62], [63], [64], [78], [65]. Yet AF is dynamic: autonomic tone, loading, inflammation, sleep, alcohol, medication, and remodeling can alter substrate over time. Prospective validation and uncertainty-aware substrate reporting remain essential before simulated drivers or vulnerable regions can be used as routine procedural targets.

Heart failure (HF) is attractive because it requires longitudinal management. Digital twins may estimate latent states such as elastance, vascular resistance, contractility, stiffness, volume status, and circulatory reserve by integrating imaging, biomarkers, renal function, device diagnostics, medication history, symptoms, and outcomes [36], [4], [5]. Potential uses include therapy titration, decompensation warning, CRT optimization, ventricular assist device planning, and remote monitoring. However, deterioration often reflects interacting triggers such as infection, arrhythmia, ischemia, renal dysfunction, nonadherence, and fluid overload, so heart-failure twins must avoid deterministic overconfidence.

Coronary artery disease (CAD) and structural heart disease offer procedural use cases. Coronary twins can integrate CT angiography, plaque, perfusion, ventricular function, and physiology to estimate ischemia, simulate flow, and test virtual revascularization [79], [80], [81], [82]. Structural heart twins may support transcatheter aortic valve replacement (TAVR), mitral repair, left atrial appendage occlusion, septal defect closure, and device sizing by simulating device–tissue interaction, obstruction, residual gradients, paravalvular leak, and flow changes [83], [84]. These models require rapid turnaround and validation against procedural and outcome endpoints, because their clinical value depends less on visual realism than on whether they change device selection, implantation strategy, complication risk, or post-procedural hemodynamics.

Congenital heart disease (CHD) is suited to digital twins because anatomy and physiology vary widely across patients. Patient-specific simulation may support surgical planning, catheter intervention, Fontan circulation assessment, shunt evaluation, and long-term surveillance [85], [86], [87], [88]. Pediatric deployment also raises special concerns about growth, sparse data, family communication, consent, and long-term consequences.

Population digital twins and *in silico* trials extend the concept from individuals to virtual cohorts. They can support device evaluation, trial design, cohort enrichment, sensitivity analysis, and regulatory evidence generation [18], [74], [89], [90]. Such studies should complement clinical evidence and must document cohort generation, coverage of physiological variability, uncertainty propagation, and external validity.

Across applications, clinical maturity remains uneven. VT ablation has the clearest near-term translational signal, AF modeling is promising but still methodologically heterogeneous, heart-failure twins may have large population impact but difficult validation endpoints, and structural or congenital applications often depend on institution-specific workflows. This uneven maturity should be reflected in claims: reviews and original studies should avoid presenting all cardiac digital

TABLE III  
MODELING STRATEGIES FOR CARDIAC DIGITAL TWINS.

| Strategy                  | Core logic                                        | Strength                                       | Limitation                                 | Best-fit uses                                          |
|---------------------------|---------------------------------------------------|------------------------------------------------|--------------------------------------------|--------------------------------------------------------|
| <b>Mechanistic models</b> | Physiological equations and numerical simulation. | Interpretable and counterfactual.              | Costly and hard to personalize.            | Electrophysiology, mechanics, therapy testing.         |
| <b>Hemodynamic models</b> | Pressure, flow, loading, valves, circulation.     | Links organ and systemic physiology.           | Boundary and latent-parameter uncertainty. | HF, valves, CHD, devices.                              |
| <b>AI models</b>          | Learn patterns from imaging, ECG, EHR, wearables. | Scalable and efficient.                        | Bias, shift, limited causality.            | Segmentation, prediction, surrogates.                  |
| <b>Hybrid models</b>      | Combine physiological constraints and learning.   | Balances fidelity, speed, and personalization. | Integration and validation complexity.     | Virtual testing, updating, uncertainty quantification. |
| <b>LLM/agent layers</b>   | Retrieve, orchestrate, summarize, audit.          | Useful for workflow and explanation.           | Hallucination and accountability risks.    | Reports, evidence tracing, interfaces.                 |

twins as equally close to deployment.

## VII. VALIDATION AND CLINICAL TRANSLATION

Trustworthy translation requires validation of the whole pipeline, not only a prediction metric. Fig. 4 outlines the staged pathway from intended-use definition to post-deployment monitoring. Technical verification asks whether segmentation, mesh generation, numerical solvers, parameter estimation, AI modules, interfaces, and visualization are implemented correctly [14], [57], [91]. Mechanistic components need solver checks, convergence analysis, boundary-condition review, reproducibility, sensitivity analysis, and version control. AI components need preprocessing consistency, calibration, robustness, input-range checks, missing-data handling, and explicit failure behavior under distribution shift.

The Cardiac Digital Twin Assurance Matrix (CDT-AM; Table V) converts this pathway into six linked assurance claims. It is an evidence-appraisal framework, not a regulatory standard. The required evidence should scale with the intended decision and its failure consequence: a research visualization and an autonomous treatment recommendation should not face the same acceptance threshold.

### A. Worked Example: VT Ablation

Consider a CDT whose intended use is to recommend candidate ablation targets for scar-dependent VT under electrophysiologist supervision. Applying CDT-AM exposes the present evidence boundary. *Data assurance* requires traceable contrast-enhanced CMR, documented scar thresholds, image-to-mesh quality control, and registration to invasive mapping. *Model assurance* requires solver checks, reproducible pacing protocols, and sensitivity to conduction and tissue-property assumptions. Current prospective evidence shows enrichment of electrogram abnormalities and identification of deceleration zones at twin-predicted sites, but does not establish unique parameter recovery [8]. *Uncertainty assurance* should present alternative circuits or targets and their dependence on scar segmentation and conduction velocity; a single deterministic lesion map is insufficient. *Clinical assurance* has reached prospective workflow feasibility: the 10-patient guided-ablation series reported acute noninducibility in all participants and freedom from sustained recurrence in eight

at a mean 13 months, but lacked a concurrent comparator [9]. *Human-system assurance* is partly addressed by supervised electrophysiologist review, while formal usability, automation-bias, and override studies remain sparse. *Lifecycle assurance*—including software-version control across centers, drift monitoring, and post-deployment incident review—has not been demonstrated. The resulting appraisal is therefore E3 for workflow feasibility, below E4 for comparative clinical benefit and E5 for monitored deployment. This worked example illustrates why a promising outcome in a small series cannot discharge all six assurance claims.

Clinical validation asks whether outputs are meaningful for the intended decision, user, population, and setting. Prediction components should follow guidance such as TRIPOD, PROBAST, and TRIPOD+AI [92], [93], [94]. Diagnostic, prognostic, and decision-support claims should be tested for discrimination, calibration, clinical utility, subgroup performance, external validity, and comparison with standard care. For procedural planning, relevant endpoints include agreement with invasive mapping, procedural decisions, lesion burden, complications, recurrence, and long-term outcomes. Prospective workflow studies and interventional trials are ultimately needed when digital twin outputs guide care [95], [96], [97], [98].

Uncertainty quantification is not optional. Input uncertainty arises from imaging, ECG, wearables, laboratories, and EHRs; parameter uncertainty from personalization; structural uncertainty from model assumptions; and numerical uncertainty from simulation [14], [99], [100]. Clinically useful systems should expose calibrated intervals or maps, alternative plausible mechanisms, sensitivity to assumptions, and conditions for abstention. A VT twin, for example, should report not only a predicted isthmus but also alternative circuits and dependence on scar thresholding or conduction velocity [101], [102].

Interpretability and usability determine whether validated models change practice. Outputs should be tailored to users: electrophysiologists need activation maps and ablation-target comparisons; heart-failure clinicians need trajectory and therapy-response estimates; structural intervention teams need device and flow simulations; patients need understandable explanations. Human-factors testing should evaluate whether the twin improves decisions without increasing cognitive burden, overreliance, or alert fatigue [103], [104].

TABLE IV  
CLINICAL APPLICATIONS MAPPED TO THE EVIDENCE LADDER. LEVELS DENOTE THE STRONGEST EVIDENCE FOR THE STATED TWIN CLAIM, NOT MATURITY OF THE CLINICAL SPECIALTY.

| Domain         | Twin function                                    | Typical inputs                       | Clinical value                           | Maturity |
|----------------|--------------------------------------------------|--------------------------------------|------------------------------------------|----------|
| VT             | Reentry and virtual-ablation planning.           | Scar CMR, CT, ECG, mapping.          | Target and lesion planning.              | E3       |
| AF             | Substrate and lesion-set simulation.             | Atrial imaging, ECG, mapping.        | Personalized rhythm-control planning.    | E2–E3    |
| HF             | Latent-state estimation and response simulation. | Echo/CMR, biomarkers, EHR, devices.  | Titration, warning, device optimization. | E1–E2    |
| CAD/structural | Flow, ischemia, and device simulation.           | CT, echo, perfusion, pressure, flow. | Procedural planning and risk estimation. | E2–E3    |
| CHD/cohorts    | Virtual surgery and population simulation.       | Imaging, catheter data, cohorts.     | Planning and evidence generation.        | E1–E2    |

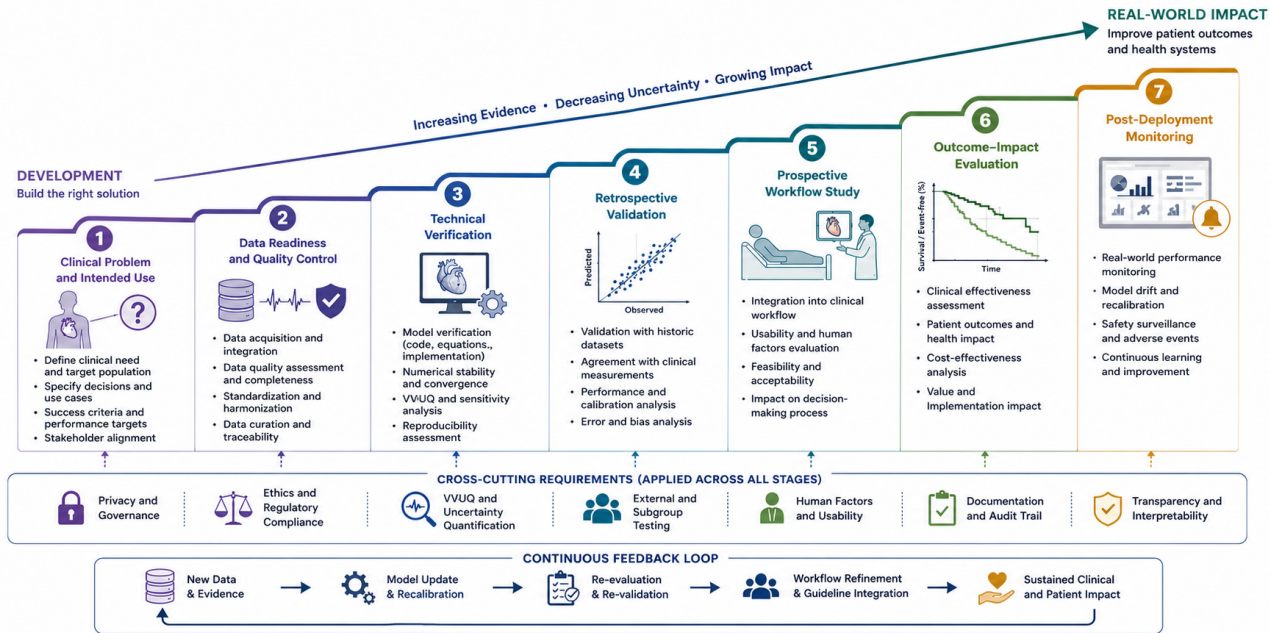


Fig. 4. Validation and translation pathway. Clinically useful cardiac digital twins require intended-use definition, data readiness, technical verification, analytical validation, clinical validation, uncertainty quantification, workflow testing, regulatory planning, and post-deployment monitoring.

TABLE V  
CARDIAC DIGITAL TWIN ASSURANCE MATRIX (CDT-AM).

| Dimension           | Assurance question                                                           | Minimum evidence                                                                   | Failure if omitted                          |
|---------------------|------------------------------------------------------------------------------|------------------------------------------------------------------------------------|---------------------------------------------|
| <b>Data</b>         | Are measurements traceable, suitable, and representative?                    | Provenance, quality control, missingness, subgroup and device reporting.           | Biased or unstable personalization.         |
| <b>Model</b>        | Is implementation correct and are parameters identifiable?                   | Code/solver verification, calibration targets, sensitivity, reproducibility.       | A plausible output for the wrong reason.    |
| <b>Uncertainty</b>  | Does output reflect input, parameter, structural, and numerical uncertainty? | Propagation, calibration, robustness, alternatives, abstention criteria.           | False precision and unsafe confidence.      |
| <b>Clinical</b>     | Is the output valid and useful for the specified decision?                   | External and prospective validation, comparator, utility and safety endpoints.     | Technical accuracy without patient benefit. |
| <b>Human system</b> | Can intended users interpret and safely act on the result?                   | Usability, automation-bias testing, equity analysis, override and accountability.  | Misuse, delay, inequity, or overreliance.   |
| <b>Lifecycle</b>    | Does performance remain controlled after change?                             | Versioning, drift/cybersecurity monitoring, controlled updates, incident response. | Silent degradation after deployment.        |

Regulation and ethics must be addressed early. A CDT that informs diagnosis, prognosis, or treatment may meet Software-as-a-Medical-Device definitions, depending on jurisdiction and intended use [105]. For AI-enabled or updated systems, pre-determined change control plans, lifecycle monitoring, and Good Machine Learning Practice provide relevant governance principles [106]–[109]. Privacy, consent, cybersecurity, access control, fairness, explainability, accountability, and equitable access must be treated as design constraints rather than post

hoc documentation [110]–[112].

The strongest evidence for clinical translation will come from prospective studies that evaluate whether digital-twin use changes management and improves outcomes. Retrospective agreement with a clinical endpoint is useful, but it does not prove improved decision quality, shorter procedures, fewer unnecessary lesions, fewer hospitalizations, or better patient-reported outcomes. Clinical-AI evaluation frameworks can help organize such studies, but CDTs additionally require

reporting of model assumptions, personalization, simulation uncertainty, and feedback updating [98].

### VIII. CHALLENGES AND FUTURE DIRECTIONS

The field should now move from impressive prototypes toward trustworthy digital twin ecosystems. Fig. 5 summarizes the roadmap from static models toward continuously updated, counterfactual, privacy-preserving, and human-supervised decision-support systems. First, cardiac twins must evolve from static baseline models to continuously updated representations. Wearables, implantable devices, repeated imaging, home measurements, laboratories, symptoms, and EHR events can recalibrate state estimates over time, but updating introduces missingness, noise, drift, version control, and governance challenges.

Second, the field must prioritize counterfactual simulation. Clinicians rarely need only a prediction; they need to compare actions. Digital twins are most valuable when they estimate plausible outcomes under alternative ablation strategies, medication titration, device settings, revascularization, or structural interventions. Counterfactual outputs must be accompanied by uncertainty and assumption sensitivity, because visually compelling simulations can create false confidence.

Third, future systems will probably be modular rather than autonomous agents. A clinical twin may combine data-quality assessment, preprocessing, model selection, simulation, uncertainty analysis, explanation, and audit. Language-model components could coordinate or summarize bounded steps, but they should not silently alter physiological parameters or treatment recommendations; source grounding, structured interfaces, versioned logs, and human approval are necessary [75]–[77].

Fourth, benchmarking and open science are needed. Studies differ widely in populations, data sources, endpoints, and assumptions, making comparison difficult. Task-specific benchmarks should be developed for VT, AF, heart failure, coronary flow, structural interventions, congenital heart disease, and population twins. Reporting should include intended use, data provenance, preprocessing, model assumptions, personalization, validation design, uncertainty, subgroup performance, computation time, workflow requirements, and post-deployment monitoring [14].

Finally, translation will depend on interdisciplinary collaboration. Cardiologists, electrophysiologists, radiologists, surgeons, biomedical engineers, computer scientists, regulators, ethicists, health-system leaders, and patients must co-design systems that are useful at real decision points. Education is also necessary: clinicians should understand model assumptions and uncertainty, while engineers should understand workflows, safety constraints, and clinical endpoints.

Two near-term priorities are particularly important. The first is benchmark construction: shared datasets and task definitions are needed for scar segmentation, ECG-to-model personalization, VT target prediction, AF recurrence modeling, heart-failure decompensation warning, and coronary or structural intervention simulation. The second is lifecycle governance: continuously updated twins must define when a model is

recalibrated, who approves changes, how performance drift is monitored, and how clinicians are notified when recommendations change. Without these foundations, cardiac digital twins may remain impressive research artifacts rather than dependable biomedical engineering systems.

### IX. LIMITATIONS OF THIS REVIEW

This is a structured state-of-the-art review rather than a registered systematic review, and it does not estimate pooled performance or treatment effect. Terminology and reporting remain heterogeneous, which makes evidence-level assignment partly interpretive. The evidence map emphasizes English-language peer-reviewed literature and authoritative guidance available through July 19, 2026; recently published, preprint-only, non-English, or poorly indexed studies may be under-represented. Evidence levels describe the strongest support for a specified claim and should not be read as a quality score for every component of a study. Finally, CDT-AM is an author synthesis intended to organize appraisal; its reliability and usefulness should be tested prospectively by independent clinical, engineering, patient, and regulatory stakeholders.

### X. CONCLUSION

Cardiac digital twins offer a powerful route toward precision cardiovascular medicine by combining multimodal patient data, mechanistic simulation, artificial intelligence, uncertainty quantification, and longitudinal updating. Their promise is clearest when they support decisions that conventional risk scores and isolated AI predictors cannot address: mechanism inference, virtual intervention testing, personalized procedural planning, therapy optimization, and dynamic follow-up. Evidence is growing in VT ablation and expanding across AF, heart failure, coronary and structural disease, congenital heart disease, remote monitoring, and in silico trials.

The decisive challenge is trustworthiness. A clinically useful CDT must demonstrate that its measurements, model, uncertainty, clinical claim, human interaction, and lifecycle controls are fit for one bounded purpose. CDT-AM and the evidence ladder make those obligations explicit. Progress should therefore be judged not by visual realism or model size, but by whether the complete data-to-decision pathway is reproducible, calibrated, safe, useful, and monitored.

### DATA AVAILABILITY

No new patient-level data were generated or analyzed for this review. All evidence is drawn from the cited literature and public regulatory or standards documents.

### ETHICS STATEMENT

This review did not involve human participants, animals, or identifiable patient data; institutional ethics approval and informed consent were therefore not applicable.

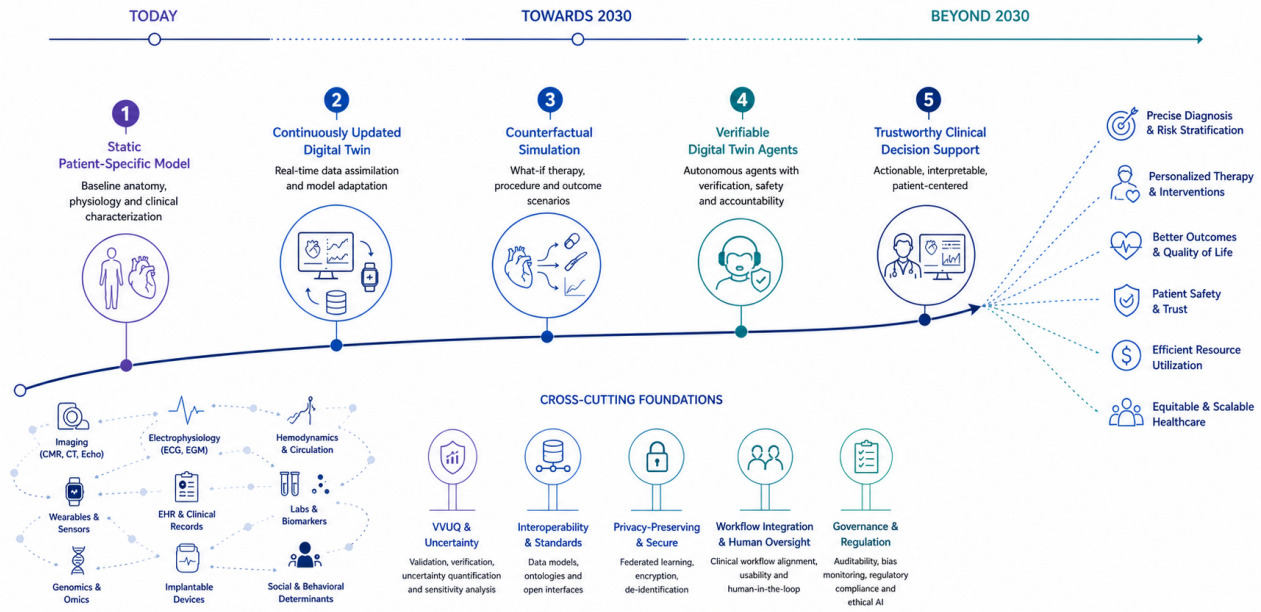


Fig. 5. Roadmap for trustworthy cardiac digital twins. Future systems should evolve from static patient-specific models toward continuously updated, counterfactual, privacy-preserving, and human-supervised decision-support ecosystems. Key enabling requirements include VVUQ, interoperability, federated deployment, benchmark datasets, reporting standards, workflow integration, and regulatory lifecycle management.

#### AUTHOR CONTRIBUTIONS

W.C.: Conceptualization, literature curation, methodology, visualization, and writing—original draft. H.Y.: Conceptualization, imaging methodology, supervision, and writing—review and editing. H.L.: Conceptualization, computational-modeling and validation oversight, supervision, and writing—review and editing.

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#### CONFLICTS OF INTEREST

The authors declare no conflicts of interest.

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